Karanveer Singh

MSc student in Molecular Biology, Umeå University. Bioinformatics and plant genomics.
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Research interests

Plant genomics and computational biology, in particular how long non-coding RNAs and gene regulation shape stress responses, studied with comparative and workflow-based genomic analysis as in my thesis on conifers. I also have experience in pharmacogenomics and structure-based drug discovery.

Education

MSc, Molecular Biology, Umeå University, Sweden — 2021 to 2022, 2026

Thesis on lncRNA in Scots pine and Norway spruce (supervisor: Nathaniel Street); advanced project on tacrolimus pharmacogenomics (supervisor: Dr. Naushad Rais, MAHE Dubai). 30 ECTS each.

B.Tech, Biotechnology, BITS Pilani Dubai Campus, UAE — 2017 to 2021

Undergraduate research in computational drug discovery and plant biochemistry (supervisor: Dr. Mainak Dutta).

Research experience

MSc thesis: lncRNA expression in Pinus sylvestris and Picea abies under abiotic stress — Jan to Jun 2026

Umeå Plant Science Centre, Umeå University. Supervisor: Nathaniel Street

- Built a Snakemake pipeline on a SLURM cluster (Micromamba environments) taking candidate lncRNAs from alignment to functional enrichment. Open source: github.com/KvS-25/comparative-lncRNA-pipeline.
- Aligned candidates to merged reference transcriptomes with minimap2, then used bedtools merge and multiinter to classify them as conserved or specific to needle, root, cold, drought and, in spruce, somatic and zygotic embryo samples.
- Filtered against curated lncRNA sets with bedtools intersect, assigned each locus its nearest protein-coding gene within 100 kb with bedtools closest, and ran GO (topGO, eggNOG-mapper) and KEGG enrichment in R, removing non-plant terms. Figures with ggplot2 and UpSetR.

MSc advanced project: CYP3A4, CYP3A5 and ABCB1 genotypes and tacrolimus exposure — May to Sep 2026

Umeå University. Supervisor: Dr. Naushad Rais, MAHE Dubai

- Systematic review and meta-analysis under PRISMA 2020: PubMed, Scopus and Embase searches, deduplication and screening in Zotero, reference mining of earlier meta-analyses, data extraction and Newcastle-Ottawa quality scoring.
- Multilevel random-effects models in R (metafor, REML) on log dose-adjusted trough concentration, with subgroup, leave-one-out and sensitivity analyses. Variant annotation with SpliceAI, Human Splicing Finder and GTEx. Open source: github.com/KvS-25/tacrolimus-pgx-meta-analysis.

B.Tech thesis: machine learning to screen for acetyl-CoA carboxylase 2 inhibitors — Jan to Jun 2021

BITS Pilani Dubai. Supervisor: Dr. Mainak Dutta

- Docked 106 literature-derived derivatives with AutoDock 4.2 and automated docking of a random quarter of DrugBank with Python scripts.
- Trained a kernel ridge regression model on Coulomb matrix representations (grid search, fivefold cross-validation) to predict binding energies for the rest of DrugBank, and checked candidates with GROMACS molecular dynamics (CHARMM36, MM-PBSA).

Lab project: repurposed drugs against the SARS-CoV-2 main protease; HPLC method development — Jan to May 2020

BITS Pilani Dubai. Supervisor: Dr. Mainak Dutta

- Docked approved drugs with AutoDock Vina in UCSF Chimera into a homology model and a crystal structure of the protease and compared interacting residues. This work fed into the publication below.
- Developed a HILIC HPLC method to separate antioxidant standards ahead of analysing Salicornia extracts.

Lab project: antioxidant activity of Salicornia grown under different salinity Aug to Dec 2019

BITS Pilani Dubai. Supervisor: Dr. Mainak Dutta

- Methanol extraction of liquid nitrogen-ground shoots, GeneVac concentration, and ABTS and DPPH radical scavenging assays on a UV/Vis spectrophotometer, expressed as Trolox equivalents.

Publication

M. N. Jayakumar, J. Pillai U., M. Mehta, **K. Singh**, E. Iype, M. Dutta. Identification of Potential Inhibitors Against SARS-CoV-2 3CLpro, PLpro, and RdRP Proteins: An In-Silico Approach. In *Advances in Computational Modeling and Simulation*, Lecture Notes in Mechanical Engineering, Springer, 2022. doi:10.1007/978-981-16-7857-8_8

Work experience

Practice School intern, GenoPhe Biotech, Bengaluru, India Jun to Jul 2019

Zebrafish toxicology and screening start-up. Reviewed zebrafish patents for a freedom-to-operate assessment and used NCBI databases, BLAST and PROMALS3D to find zebrafish homologs of human cardiac development and heart disease genes.

Skills

Programming and workflows: R, Python, Bash and Linux, Snakemake, SLURM and HPC, Micromamba, Git and GitHub

Sequence and genomics analysis: minimap2, bedtools, BLAST, PROMALS3D, eggNOG-mapper, GO and KEGG enrichment (topGO)

Statistics and evidence synthesis: multilevel random-effects meta-analysis (metafor), PRISMA 2020, Newcastle-Ottawa Scale, Zotero

Variant annotation: SpliceAI, Human Splicing Finder, GTEx

Computational drug discovery: AutoDock 4 and Vina, GROMACS and MM-PBSA, UCSF Chimera and ChimeraX, Discovery Studio, Open Babel, kernel ridge regression

Laboratory: HPLC method development, ABTS and DPPH antioxidant assays, UV/Vis spectrophotometry, plant extraction

Visualisation: ggplot2, UpSetR